

Real Business Cycle (RBC)

1. □ Set up consumer's problem in RBC model and explain it.
- The basic RBC model is simply the Newclassical Growth Model (NGM) in discrete time with two additions:
 - i. productivity shocks and ii. labor supply choice.

□ Consumer's problem:

$$\text{Max } E \left[\sum \beta^t \left((1-\phi) u(c_t) + \phi v(h_t) \right) \right]$$

s.t.

- i) budget constraint: $K_{t+1} = l_t w_t + (1+r_t) K_t - c_t$
- ii) production function: $f(K_t, l_t, Z_t) = Z_t K_t^\alpha l_t^{1-\alpha}$
- iii) labor endowment: $h_t + l_t = 1 \Rightarrow l_t = 1 - h_t$
- iv) productivity shock process: $Z_{t+1} = \phi Z_t + \varepsilon_{t+1}$

Here,

$u(c_t)$ = utility from consumption

$v(h_t)$ = utility from leisure

Two control variables – consumption (c_t) and leisure (h_t)

State variable K_t is the capital stock.

r_t = rate of return on capital (net of depreciation).

l_t = amount of labor

Z_t = productivity parameter

2. □ Set up the Lagrangian and find the optimality condition.

□ The Lagrangian:

$$L = E \left[\sum \beta^t (1-\phi) u(c_t) + \phi v(h_t) \right] + \lambda_t \left((1-h_t)w_t + (1+r_t)K_t - c_t - K_{t+1} \right)$$

□ The optimality condition (FOC):

$$\begin{aligned} \cdot \frac{\partial L}{\partial c_t} = 0 &\Rightarrow \beta^t (1-\phi) u'(c_t) - \lambda_t = 0 \\ &\Rightarrow \beta^t (1-\phi) u'(c_t) = \lambda_t \quad \text{----- (1)} \end{aligned}$$

Updating,

$$\frac{\partial L}{\partial c_{t+1}} \Rightarrow E \beta^{t+1} (1-\phi) u'(c_{t+1}) = \lambda_{t+1} \quad \text{----- (1a)}$$

$$\begin{aligned} \cdot \frac{\partial L}{\partial h_t} = 0 &\Rightarrow \beta^t \phi v'(h_t) - \lambda_t w_t = 0 \\ &\Rightarrow \beta^t \phi v'(h_t) = \lambda_t w_t \quad \text{----- (2)} \end{aligned}$$

$$\begin{aligned} \cdot \frac{\partial L}{\partial K_{t+1}} &= + \lambda_{t+1} (1+r_{t+1}) = + \lambda_t \\ &\Rightarrow \lambda_t = \lambda_{t+1} (1+r_{t+1}) \quad \text{----- (3)} \end{aligned}$$

3. ☐ Find the Euler Equation and real wage, and interpret them.
Find the intra-period choice of c_t and h_t , and interpret it.

Euler Equation:

From equation (1) $\Rightarrow$

$$\beta^t (1-\phi) u'(c_t) = \lambda_t$$

$$\Rightarrow \beta^t (1-\phi) u'(c_t) = \lambda_{t+1} (1+r_{t+1}) \quad [\text{from (3)}]$$

$$\Rightarrow \beta^t (1-\phi) u'(c_t) = E \beta^{t+1} (1-\phi) u'(c_{t+1}) (1+r_{t+1}) \quad \left[\begin{array}{l} \text{from (1a) value} \\ \text{of } \lambda_{t+1} \end{array} \right]$$

$$\Rightarrow u'(c_t) = \beta E u'(c_{t+1}) (1+r_{t+1}) \quad \text{----- (4)}$$

This is our Euler equation. It shows the intertemporal choice of consumption.

Here,

$u'(c_t)$ = MU of today.

$\beta E u'(c_{t+1})$ = Discounted expected MU of tomorrow.

$(1+r_{t+1})$ = Gross rate of return on k_{t+1} .

□ Intra-period choice of consumption and leisure:

From ② $\Rightarrow$

$$\bullet \quad \beta^t \phi v'(h_t) = \lambda_t w_t$$

$$\Rightarrow \beta^t \phi v'(h_t) = \beta^t (1-\phi) u'(c_t) w_t \quad [\text{from ① value of } \lambda_t]$$

$$\Rightarrow \phi v'(h_t) = (1-\phi) u'(c_t) w_t \quad \text{----- ⑤}$$

This is our intra-period choice of consumption (c_t) and leisure (h_t). It shows that the marginal benefit of additional consumption equals the marginal cost of lost leisure.

4. □ Show the SR and LR labor-leisure trade-off applying log utility function.

□ SR labor supply:

The log utility function to see the labor-leisure trade-off (i.e., relationship between labor hours and wage rate) is:

$$u(c_t, h_t) = (1-\phi) \log c_t + \phi \log h_t$$

s.t., budget constraint: $K_{t+1} = l_t w_t + (1+r_t)K_t - c_t$

labor endowment: $h_t + l_t = 1 \Rightarrow l_t = 1 - h_t$

Lagrangian:

$$L = E \left[\sum \beta^t \left((1-\phi) \log c_t + \phi \log h_t \right) \right] + \lambda_t \left((1-h_t)w_t + (1+r_t)K_t - c_t - K_{t+1} \right)$$

□ FOC:

$$\begin{aligned} \cdot \frac{\partial L}{\partial c_t} = 0 &\Rightarrow \frac{\beta^t (1-\phi)}{c_t} - \lambda_t = 0 \\ &\Rightarrow \frac{\beta^t (1-\phi)}{c_t} = \lambda_t \end{aligned} \quad \text{----- (1)}$$

updating,

$$\frac{\partial L}{\partial c_{t+1}} \Rightarrow \frac{\beta^{t+1} (1-\phi)}{c_{t+1}} = \lambda_{t+1} \quad \text{----- (1a)}$$

$$\begin{aligned} \cdot \frac{\partial L}{\partial h_t} = 0 &\Rightarrow \frac{\beta^t \phi}{h_t} - \lambda_t w_t = 0 \\ &\Rightarrow \frac{\beta^t \phi}{h_t} = \lambda_t w_t \end{aligned} \quad \text{----- (2)}$$

$$\cdot \frac{\partial L}{\partial K_{t+1}} = 0 \Rightarrow \lambda_t = \lambda_{t+1} (1+r_{t+1}) \quad \text{----- (3)}$$

Euler Equation:

from (2) $\Rightarrow$

$$\bullet \frac{\beta^t \phi}{h_t} = \lambda_t w_t$$

$$\Rightarrow \frac{\beta^t \phi}{h_t} = \frac{\beta^t (1-\phi) w_t}{c_t} \quad [\text{value of } \lambda_t \text{ from equation (1)}]$$

$$\Rightarrow \phi c_t = h_t (1-\phi) w_t \quad \text{----- (4)}$$

$$\Rightarrow \phi c_{t+1} = h_{t+1} (1-\phi) w_{t+1} \quad (\text{updated})$$

Now, from (1) $\Rightarrow$

$$\bullet \frac{\beta^t (1-\phi)}{c_t} = \lambda_t$$

$$\Rightarrow \frac{\beta^t (1-\phi)}{c_t} = \lambda_{t+1} (1+r_{t+1}) \quad [\text{Substituting for } \lambda_t \text{ from (3)}]$$

$$\Rightarrow \frac{\beta^t (1-\phi)}{c_t} = \frac{\beta^{t+1} (1-\phi)}{c_{t+1}} (1+r_{t+1}) \quad [\text{Substituting for } \lambda_{t+1} \text{ from (1a)}]$$

$$\Rightarrow \frac{1}{c_t} = \frac{\beta (1+r_{t+1})}{c_{t+1}} \quad [\text{Euler equation}]$$

$$\Rightarrow \frac{\phi}{h_t (1-\phi) w_t} = \frac{\beta (1+r_{t+1}) \phi}{h_{t+1} (1-\phi) w_{t+1}} \quad [\text{from equation (4)}]$$

$$\Rightarrow \frac{h_{t+1}}{h_t} = \beta (1+r) \frac{w_t}{w_{t+1}} \quad \left[\begin{array}{l} \text{Assuming } r \text{ is exogenous} \\ \therefore r_{t+1} = r_t = r \end{array} \right]$$

This says that leisure hour decreases if wage increases in the next period and leisure is relatively high when wage rate is relatively low. This establishes positive relation between w_t and c_t in the SR.

□ LR labor supply:

Assuming wages and income are constant, since consumption is constant at income level:

$$\cdot C_t = w h_t \cdot$$

$$\Rightarrow C_t = w (1 - h_t)$$

----- (5)

from (4) $\Rightarrow$

$$\cdot \phi C_t = h_t (1 - \phi) w$$

$$\Rightarrow \phi w (1 - h_t) = h_t (1 - \phi) w \quad \left[\text{Substituting for } C_t \text{ from (5)} \right]$$

$$\Rightarrow \phi - \phi h_t = h_t - \phi h_t$$

$$\therefore h_t = \phi$$

This says that leisure is independent of the wage level in the long run. It does not mean labor hour is constant, but relationship between leisure and wage breaks down.

Leisure/labor can be affected by other things i.e., change in interest rate.

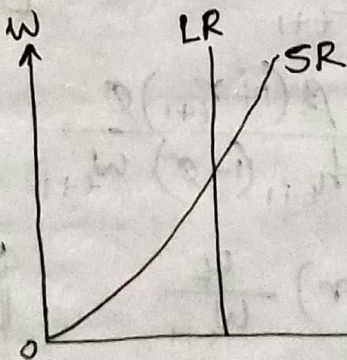


fig: labor supply in SR and LR.

While the long-run elasticity of labor supply may be zero, it is positive in the short run with respect to wage.

5. □ How does a real shock generate business cycle? Show with elasticity of labor supply.

□ How real shock generates business cycle:

Considering — a positive productivity shock making factors of production more productive. As a result, both wage ($w = MP_L$) and interest rates ($r = MP_K$) will rise.

• How labor supply responds to this shock is crucial to generate function fluctuation in real output. This responsiveness is measured by elasticity of labor supply. If elasticity of labor supply is very high, then a change in wage (following a real shock) will bring large variation in employment and output. That will be sufficient to match real data.

□ Finding elasticity of labor supply:

To calculate the elasticity of labor supply, let's ^{consider} ~~assume an~~ ~~utility function from leisure~~ the following constant relative risk aversion (CRRA) utility function:

$$V(h_t) = \frac{\sigma}{1-\sigma} h_t^{\frac{\sigma-1}{\sigma}} \quad \text{CRRA: } U(c_t, h_t) = \log c_t - \frac{\sigma}{1-\sigma} h_t^{\frac{\sigma-1}{\sigma}}$$

$$\therefore V'(h_t) = h_t^{-\frac{1}{\sigma}}$$

Now, from labor market equilibrium:

$$\phi V'(h_t) = (1-\phi) U'(c_t) w_t$$

$$\Rightarrow \phi h_t^{-\frac{1}{\sigma}} = \lambda w_t$$

$$\Rightarrow h_t = \left(\frac{\lambda w_t}{\phi} \right)^{-\sigma}$$

$$\Rightarrow 1 - l_t = \left(\frac{\lambda w_t}{\phi} \right)^{-\sigma}$$

$$\therefore l_t = 1 - \left(\frac{\lambda w_t}{\phi} \right)^{-\sigma}$$

$\left[\lambda \left((1-\phi) U'(c_t) \right) \text{ is the constant marginal utility of consumption} \right]$

$$[\because h_t + l_t = 1]$$

①

Now, the labor supply elasticity (in short-run) is :

$$\varepsilon_{L,W} = \frac{dL}{dW} \cdot \frac{W}{L}$$

$$= \sigma \left(\frac{\lambda W}{\phi} \right)^{-\sigma-1} \cdot \frac{\lambda}{\phi} \times \frac{W}{L}$$

$$= \sigma \left(\frac{\lambda W}{\phi} \right)^{-\sigma} \cdot \frac{1}{L}$$

$$= \sigma \frac{h}{L}$$

$$\left[\because h = \left(\frac{\lambda W}{\phi} \right)^{-\sigma} \text{ from } \right]$$

$$\therefore \varepsilon_{L,W} = \frac{h}{L}$$

$$\left[\text{Assuming } \sigma = 1 \text{ (logarithmic utility in leisure)} \right]$$

Further assuming, ϕ and λ are such that $\frac{h}{L} = 2$ (thinking about an 8-hour workday). Here, $\varepsilon_{L,W} = 2$ does not seem to be enough to replicate the employment fluctuations observed in the data.

On the other hand, $\varepsilon_{L,W} = 2$ seems very large as compared with micro data. An individual does not reduce 20% working hours ~~falling~~ following a 10% wage reduction.

6. ☐ What did Hansen propose about the labor supply elasticity?

☐ The indivisible labour solution:

Hansen proposed that labor large variation in labor supply comes from a wage change because, labor is indivisible and hence a worker responds by not to work at all (extensive margin). This implies aggregate elasticity is large even though individual elasticity is low. Now the challenge is to model such a problem.

• Hansen models by assuming that —

i. there are fixed costs of going to work. This makes labor supply very responsive for a range of wage levels. The decision variables are now both — days of work: $d \leq \bar{d}$, and — the hours of work each day: n

ii. there is a fixed community cost: m , (in terms of utility) which one pays if she decides to work on that day, regardless of how many hours she works.

∴ The objective function:

$$\text{Max } E \left[\sum \beta^t (u(c_t) - d_t v(n_t) - m_t d_t) \right]$$

$$\text{S.t.}, K_{t+1} = n_t d_t w_t + (1+r_t) K_t - c_t$$

$$\begin{cases} n_t = \text{hours worked} \\ d_t = \text{days worked} \\ n_t d_t w_t = \text{wage income} \\ m_t d_t = \text{per day community cost} \end{cases}$$

∴ The Lagrangian is:

$$L = E \left[\sum \beta^t (u(c_t) - d_t v(n_t) - m_t d_t) \right] + \lambda_t (n_t d_t w_t + (1+r_t) K_t - c_t - K_{t+1})$$

FOC:

$$\begin{aligned} \text{i) } \frac{\partial L}{\partial c_t} = 0 &\Rightarrow \beta^t u'(c_t) - \lambda_t = 0 \\ &\Rightarrow \beta^t u'(c_t) = \lambda_t \text{ ----- } \textcircled{1} \\ &\Rightarrow \beta^{t+1} u'(c_{t+1}) = \lambda_{t+1} \text{ (Updating)} \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{\partial L}{\partial n_t} = 0 &\Rightarrow -\beta^t d_t v'(n_t) + \lambda_t d_t w_t = 0 \\ &\Rightarrow \beta^t v'(n_t) = \lambda_t w_t \text{ ----- } \textcircled{2} \end{aligned}$$

$$\begin{aligned} \text{iii) } \frac{\partial L}{\partial K_{t+1}} = 0 &\Rightarrow -\lambda_t + \lambda_{t+1} (1+r_{t+1}) = 0 \\ &\Rightarrow \lambda_t = \lambda_{t+1} (1+r_{t+1}) \text{ ----- } \textcircled{3} \end{aligned}$$

$$\begin{aligned} \text{iv) } \frac{\partial L}{\partial d_t} &\Rightarrow -\beta^t v(n_t) - \beta^t m_t + \lambda_t n_t w_t \leq 0 \\ &\Rightarrow \beta^t v(n_t) + \beta^t m_t \geq \lambda_t n_t w_t \text{ ----- } \textcircled{4} \end{aligned}$$

Euler Equation:

$$\begin{aligned} \text{from } \textcircled{1} &\Rightarrow \beta^t u'(c_t) = \lambda_t \\ &\Rightarrow \beta^t u'(c_t) = \lambda_{t+1} (1+r_{t+1}) \quad [\text{from } \textcircled{3}] \\ &\Rightarrow \beta^t u'(c_t) = \beta^{t+1} u'(c_{t+1}) (1+r_{t+1}) \quad [\text{from } \textcircled{1}] \\ &\Rightarrow u'(c_t) = \beta u'(c_{t+1}) (1+r_{t+1}) \text{ ----- } \textcircled{5} \end{aligned}$$

This is our Euler equation.

$$\text{from ④} \Rightarrow \beta^t v(n_t) + \beta^t m_t \geq \lambda_t n_t w_t$$

$$\Rightarrow \beta^t (v(n_t) + m_t) \geq \beta^t u(c_t) n_t w_t$$

$$\Rightarrow v(n_t) + m_t \geq u(c_t) n_t w_t \quad \text{----- ⑥}$$

Here, L.H.S. = cost if one works

R.H.S. = benefit if she works.

This says one will not participate if cost outweighs the benefit. This implies there is a reservation wage below which this representative worker will not participate. A slight increase in market wage will cause huge participation. Therefore, at this reservation wage level labor supply is horizontal.

$$\text{from ②} \Rightarrow \beta^t v'(n_t) = \lambda_t w_t$$

$$\Rightarrow \beta^t v'(n_t) = \beta^t u'(c_t) w_t$$

$$\Rightarrow v'(n_t) = u'(c_t) w_t$$

$$\Rightarrow v'(n_t) = \lambda w_t$$

[Assuming $\beta(1+r)=1$, c_t becomes constant even time. $\therefore u'(c_t) = \lambda_t = \lambda$]

This implicitly defines n_t as a function of real wage.
i.e., $n_t = n^*(w)$

positively sloped segment of the labor supply curve.

$$\text{from ⑥} \Rightarrow v(n_t) + m_t \geq u(c_t) n_t w_t$$

$$\Rightarrow v(n^*) + m_t \geq \lambda n^* w_t \quad [\text{Assuming } c_t \text{ is constant}]$$

if $v(n^*) + m > \lambda n^* w$, then $d=0$, otherwise $d=\bar{d}$.

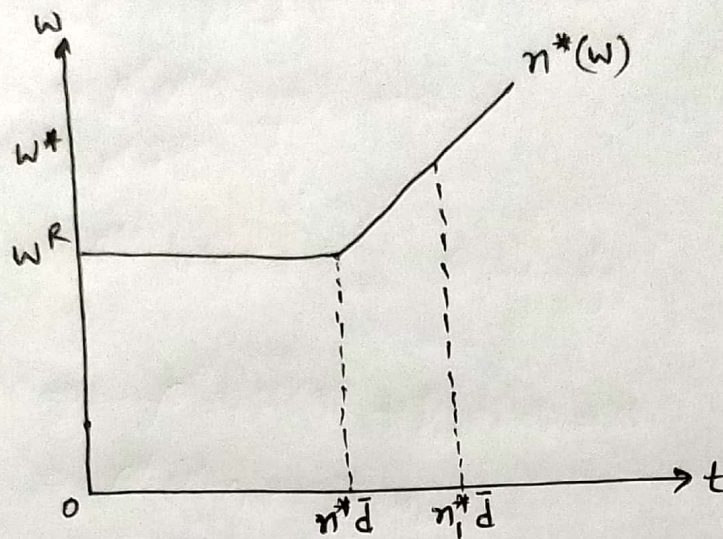


Fig: The Hansen Labour supply.

- At $w = w^R$, total labor supply is n^*d (if he chooses to work).
- At a higher wage ($w^* > w^R$), working hours will be higher, say n_1^* per day, which gives total working hours n_1^*d amount.

⇒ However, even if individual elasticity is low, aggregate labor supply elasticity is quite large. This is what we needed to match the data.